

Foundations of C Programming (ST-1B)

1. Introduction

C is a procedural programming language created by **Dennis Ritchie**.

Used for:

- System programming
 - Embedded systems
 - OS development
-

2. Structure of C Program

```
#include<stdio.h>
```

```
int main()  
{  
    printf("Hello World");  
    return 0;  
}
```

3. Variables and Data Types

Type	Example
------	---------

int	10
-----	----

float	3.14
-------	------

char	'A'
------	-----

Example:

```
int age=20;  
float marks=89.5;
```

```
char grade='A';
```

4. Operators

Arithmetic

+ - * /

Relational

> < >= <= == !=

Logical

&& || !

5. Conditional Statements

```
if(a>10)
printf("Greater");
else
printf("Smaller");
```

6. Loops

For loop

```
for(int i=1;i<=5;i++)
{
printf("%d",i);
}
```

While loop

```
while(i<10)
{
i++;
}
```